

CROWD DENSITY ESTIMATION USING STEREO VISION

OBJECTIVE:

The system aims to classify crowd density into **Low, Medium, or High** levels by analyzing video input, and further enhance accuracy using **stereo vision–based depth estimation**.

This project focuses on:

- Improving crowd monitoring
- Supporting public safety and surveillance
- Providing scalable and real-time crowd analysis

PROJECT DEFINITION:

Crowd Density Estimation (CDE) is the process of determining how crowded a particular area is by analyzing visual data such as images or videos.

In this project:

- **Monocular (single-camera) vision** is used as the baseline method
- **Stereo vision (dual-camera)** is used as an advanced method to incorporate depth information
- Crowd density is dynamically estimated based on detected people and spatial coverage

PROPOSED SOLUTION:

The system processes video input to:

1. Detect people in each frame
2. Estimate crowd density dynamically
3. Classify density as Low / Medium / High
4. Visualize results through a web-based UI

Two methods are implemented and compared:

- Method 1: Monocular (Single Camera)
- Method 2: Stereo Vision (Depth-based)

TECHNOLOGIES USED

Frontend : HTML5,CSS3, JavaScript

Backend : Python,Flask

Computer Vision & ML : OpenCV , YOLOv4-Tiny (Object Detection), Stereo Matching Algorithms (Disparity & Depth Maps)

Hardware (Phase 2)

- Stereo Camera
- JetsonOrin Nano Super Developer Kit

METHODOLOGY:

Phase 1: Monocular Crowd Density Estimation

Input:

- Live camera feed or video file

Steps:

1. Capture video frames
2. Detect people using YOLOv4-Tiny
3. Count detected persons
4. Calculate frame coverage
5. Classify density:
 - LOW
 - MEDIUM
 - HIGH

PHASE 2: STEREO VISION–BASED CROWD DENSITY ESTIMATION

Phase 2 focuses on enhancing crowd density estimation by incorporating stereo vision to obtain depth information from the scene. In this phase, synchronized stereo image pairs are used to generate disparity and depth maps, which help estimate the distance of individuals from the camera. This depth information is combined with object detection results to improve the accuracy and reliability of crowd density estimation, particularly in scenarios with overlapping or occluded people. The enhanced system is intended for real-time deployment on edge hardware such as the Jetson Orin Nano.

CURRENT PHASE: MODEL ADAPTATION AND DEPLOYMENT PREPARATION

The current phase involves preparing the existing system for stereo-based deployment. The CNN model, which was originally trained on RGB images, is being adapted to support RGB–Depth data through retraining. Additionally, the software environment is being configured to use Python 3.8 to ensure compatibility with the Jetson Orin Nano platform. These steps are essential to enable seamless integration of depth information and support efficient real-time execution on edge devices.

CHALLENGES FACED:

- Stereo camera hardware is not yet available, limiting real-time testing and validation of depth-based crowd density estimation.
- Lack of publicly available stereo crowd image datasets, making training and evaluation of RGB–Depth models challenging.
- Current development relies on monocular data and simulated or generated stereo inputs.
- Target deployment platform Jetson Orin Nano supports Python 3.8, while the existing codebase was developed using Python 3.12, requiring environment and dependency adjustments.

OUTPUT:

METHOD -1 (MONOCULAR APPROACH)



